A family of defining quartics in 279 variables

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Abstract

Under the stated arithmetic premises of the elliptic source, we specify an integer quartic family in 279 total variables whose sufficiently large members define the integers in the rationals. Four disjoint translates of the contact set give a height exponent 19/4. This permits quadratic Taylor approximations to the formal logarithm, with comparison orders twelve and three. Even valuations of reciprocal abscissas allow sixteen local tests to be represented by five factors. The complete quadratic and expanded coefficient tables are independently verified. The family has 3692 monomials for every positive parameter, and exact coefficient height $80989^{4N+2}+1$ for $N \geq 23$. The sufficient defining parameter remains ineffective because the aggregate-trace height theorem has an ineffective additive constant.

1 The family and its arithmetic scope

All variable counts include the distinguished input X. For a fixed integer $N \geq 1$ let $A_N = 80989^{2N+1}$. The complete tables `quadratic-family279.json` and `polynomial-family279.json` specify 253 primitive integer quadratic polynomials and their sum of squares

$$P_N(X, Y_1, \ldots, Y_{278}) = \sum_{j=1}^{253} Q_{N,j}^2.$$

The parameter A_N is a coefficient, not a variable.

Theorem 1.1. *Assume the arithmetic premises stated below. There is a fixed integer $N_0 \geq 27464$ such that every $N \geq N_0$ satisfies*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{278}\ P_N(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

For every $N \geq 1$ the polynomial P_N has degree exactly four, content one, and 3692 nonzero coordinate monomials. For $N \geq 23$,

$$H(P_N) = 80989^{4N+2} + 1.$$

The construction specifies every member exactly. The proof does not determine a sufficient numerical value of N_0.

Set

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad D = (3-\rho)^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$
$$P = 2(1+\rho, 2\rho) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

We retain the following arithmetic premises, with the precise domains and hypotheses recorded in [1, 2, 3]:

1. $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n, together with the specified point charts and group domains.

2. The five-contact height estimate with exponent nineteen, exceptional cutoff 76176, and the fixed constant
$$H = 2^{110}210^{1800}\left(\prod_{p\le 76176} p\right)^{19} + 128.$$

3. The pole obstruction outside $2, 3, 17$, and its simultaneous realization for positive inputs less than 17 with square reduced denominator and the stated local squareclasses and valuation bounds at those three primes. Its normalized quadratic equations are retained.

4. The simultaneous prime-pattern, conjugate reduction-order, finite congruence, and real approximation assertions used by the source. The weighted selection and fixed denominator dilation are justified from the cited analytic estimates in [7].

The aggregate trace-height theorem used below is the proved consequence of the family height theorem described in [6]; its additive constant is ineffective. The contact amplification in [5] is an elementary consequence of the retained five-contact estimate, not an additional arithmetic assumption. Exact coefficient verification does not prove the retained arithmetic premises.

2 Four contact blocks and the two scales

Put
$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad \Theta_4 = \bigcup_{j=0}^{3}(\Theta + 3j), \qquad F_4(t) = \prod_{\theta\in\Theta_4}(t-\theta).$$

Let Σ consist of the primes at most 76176 and the two primes $123121, 161303$. Outside this set, all contact differences and all numbers $1-\theta$ are units. The contact radical $M_4(t)$ is the product of primes $p \notin \Sigma$ for which $v_p(t-\theta)$ is positive and odd for some $\theta \in \Theta_4$. The exact contact certificate in [5] verifies this exceptional set, including the normalizing value $t = 1$.

Let $E_4 = 123121 \cdot 161303 = 19859786663$. The contact amplification theorem gives
$$h(t) \le \log 26 + HE_4^{19}M_4(t)^{19/4} \le H_4M_4(t)^{19/4}, \qquad H_4 = \log 26 + HE_4^{19}, \tag{1}$$
for $t \notin \Theta_4$, and $M_4(1) = 1$. For clarity, its proof applies the five-contact estimate to the four maps $2 + 1/(t-3j)$, each with determinant -1. Their contact preimages are disjoint outside Σ. Each map changes logarithmic height by at most $\log 26$. Multiplication of the four resulting inequalities and disjointness of the radicals gives (1). The full treatment of fixed exceptional primes is in [5].

The retained root points are
$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|),$$
with nonzero integer labels. Set
$$T = \frac{1}{2}\operatorname{Tr}_{K/\mathbb{Q}}\left(\sum_{i=1}^{4} x(Q_i)\right).$$
It is positive, and the aggregate trace theorem gives fixed constants $M_1 > 0$, $M_2 \ge 1$ such that
$$B^2 \le M_1h(T) + M_2. \tag{2}$$
Use the two nonzero rational square scales $a_* = 1, T^2$. For each scale define, on its nonzero domain,
$$t_* = a_*\lambda, \quad \delta = t_* - 1, \quad R = \frac{F_4(t_*)}{a_*^{20}\delta^{22}}, \quad e = \frac{(a_*\delta)^{20}}{F_4(t_*)}, \quad q = \frac{1}{R\delta^4}. \tag{3}$$

The actual quadratic system clears these expressions by

$$F_4(t_*)e = (a_*\delta)^{20}, \qquad q\delta^2 = e. \tag{4}$$

In particular $e\delta^2 = R^{-1}$ when $\delta \neq 0$. No contact value can occur, since $F_4(t_*) = 0$ would contradict the first equation. If $q = 0$, then $e = 0$, and that equation forces $\delta = 0$. At $\delta = 0$ the contact radical is one; no reciprocal is used on this stratum.

If both radicals divide a nonzero integer J to order k, then $2h(T) \leq h(\lambda) + h(T^2\lambda)$ and (1)–(2) imply

$$B^2 \leq C|J|^{19/(4k)} + M_2, \qquad C = M_1H_4. \tag{5}$$

The two radicals need not be coprime for this implication.

3 Local tests and the approximation denominators

Let σ be conjugation, $f = 1/x$, and $z = -x/y$, extended by zero at the identity where these functions are regular. The retained point relations are

$$U_0 = H, \quad V_0 = H - P, \quad U_i = [u_i]P, \quad V_i = U_i - Q_i.$$

The charts permit $U_i = O$ or $V_i = O$ for $i > 0$. Put

$$G_2(w) = 1 - \frac{2}{5}w - \frac{2}{15}w^2, \quad \ell_0 = z(V_0)G_2(Df(V_0)^2), \quad L_i = \frac{z(U_i)}{z(H)},$$

$$L_1' = \sigma\left(\frac{z(V_1)}{\ell_0}\right), \qquad L_i' = \sigma\left(\frac{z(V_i)G_2(Df(V_i)^2)}{\ell_0}\right) \quad (i > 1).$$

These quotients are interpreted only after the denominator exclusions below. Write

$$D_H = f(H), \quad A_i = f(U_i), \quad B_i = \sigma f(V_i), \quad C_i = L_i - L_i', \quad I = L_1 - \beta, \quad \beta = 51X^2 + 2504,$$
$$G = L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1.$$

The required orders are six for D_H, twelve for C_2, C_3, C_4, G, three for C_1, I, and one for the nine A_i, B_i.

Lemma 3.1. *The polynomial G_2 has no zero in K.*

Proof. Its roots are $(-3 \pm \sqrt{39})/2$, and 39 is neither a rational square nor twice a rational square. Explicitly, if $(u + v\sqrt{2})^2 = 39$ with $u, v \in \mathbb{Q}$, then $uv = 0$; the two cases would require $u^2 = 39$ or $v^2 = 39/2$, both impossible. The independent finite certificate also verifies that the integer numerator $15 - 6w - 2w^2$ has no root modulo 17, a prime that splits in K. Its leading coefficient is a unit there, so a hypothetical local root would be integral and give a forbidden residue root. □

The finite H chart is $xf = 1$, $S^2 = x - Df$, $H = (x, xS)$. The source group calculation writes $H = [m]R_0 + T_0$ with $P = 2R_0$, $m \in \mathbb{Z}$, and $T_0 \in E[2](K)$. Put $h = m/2$ for its rational free index. The exceptional classes $E[2](K), \pm P + E[2](K)$ are excluded before any logarithmic ratio is interpreted. Indeed their finite allowed reciprocal abscissas are units outside the fixed small primes, whereas extraction below would require positive valuation at any contact. All radicals would then be one, forcing $B^2 \leq C + M_2$ by (5). The retained scalar lower bound in Section 5 contradicts this for every sufficiently large parameter. The explicit torsion translations and norm values supporting the unit assertion are those recorded in [4, Section 3]. This exclusion uses no comparison quotient.

Consequently $m \neq 0, 2$, and H, V_0 are nontorsion. Their required nonzero abscissas and ordinates, and all subtraction denominators, follow from the retained chart identities. In particular $z(H), z(V_0) \neq 0$. Lemma 3.1 gives $\ell_0 \neq 0$ globally. The analytic formal logarithms have a separate nonvanishing proof from the formal logarithm isometry when the points are formal.

4 Five factors and extraction at one place

At a contact prime $p \notin \Sigma$, δ is a unit and $\nu = v_p(R)$ is odd; it need not initially be positive. All valuations of K are unramified and normalized by $v(p) = 1$. For a point on E, every finite value of $v(f)$ is even: if $v(x) < 0$ then $2v(y) = 3v(x)$, while if $v(x) > 0$ then $2v(y) = v(x)$; the remaining case is valuation zero. This also holds for conjugated reciprocal abscissas.

Use

$$\begin{aligned}
T_1 &= 1 + q^{23}C_2^2 + q^{10}D_H^2 + q^4A_3^4 + q^8A_4^8,\\
T_2 &= 1 + q^{23}C_3^2 + q^{10}C_1^4 + q^4B_0^4,\\
T_3 &= 1 + q^{23}C_4^2 + q^2A_1^2 + q^4B_2^4,\\
T_4 &= 1 + q^{23}G^2 + q^2A_2^2 + q^4B_3^4,\\
T_5 &= 1 + q^5I^2 + q^2B_1^2 + q^4B_4^4, \qquad W = T_1T_2T_3T_4T_5.
\end{aligned}$$

Proposition 4.1. *At each contact prime, the retained pole system and the new cleared trace equations imply $\nu > 0$ and give all sixteen required test orders at one place above that prime.*

Proof. In every factor the finite nonconstant valuations in successive positions have exact powers of two divisors $1, 2, 4, 8$, omitting the last when absent. For example the first factor has valuations

$$2v(C_2) - 23\nu, \quad 2(v(D_H) - 5\nu), \quad 4(v(A_3) - \nu), \quad 8(v(A_4) - \nu).$$

Each parenthesis that is required to be odd is odd, by evenness of the reciprocal-abscissa valuations. The second position involving C_1 is $2(2v(C_1) - 5\nu)$ and is likewise twice an odd integer. The first position of T_5 is $2v(I) - 5\nu$, which is odd. Thus the minimum, including the constant term of valuation zero, is attained once. Every factor is nonzero and has valuation at most zero, and a factor of valuation zero is one in the residue field.

Write $W = w_0 + w_1\rho$. The literal cleared trace equations are

$$w_0^2 - 2w_0\tau - 2w_1^2 = 0, \qquad e\delta^2\mathfrak{b} = \tau^2.$$

Since $W \neq 0$, the first equation gives $w_0 \neq 0$ and $\tau = 1/\operatorname{Tr}(W^{-1}) \neq 0$. Therefore $\mathfrak{b} = R/\operatorname{Tr}(W^{-1})^2$. If $\ell = \max_{v|p} v(W) \leq 0$, the trace inequality gives $v_p(\mathfrak{b}) \leq \nu + 2\ell$. This is an odd valuation; the pole obstruction forces it positive. Hence $\nu > 0$ and $\ell > -\nu/2$. At a place attaining ℓ, every term of every factor has valuation strictly greater than $-\nu/2$, because the factor valuations are nonpositive and add to ℓ.

The first-column comparisons satisfy $v(C) > (23/2 - 1/4)\nu \geq 11.25$, hence order twelve. For I, the bound is $v(I) > (5/2 - 1/4)\nu \geq 2.25$, hence order three. The C_1 term gives $v(C_1) > (10/4 - 1/8)\nu \geq 2.375$, also order three. For D_H it gives $v(D_H) > (5 - 1/4)\nu \geq 4.75$; evenness makes this at least six. Every order-one term gives $v(f) > (1 - 2^{-j-1})\nu > 0$ in its position $j \geq 1$. Zero tests automatically satisfy the asserted inequalities. □

4.1 Integrality precedes the depth estimate

The extracted positive f valuations put H, U_i on the primary formal branch and V_0, V_i on the conjugate formal branch. At a nonidentity formal point of depth $r = v(z) \geq 1$, write

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) = G(Df(Q)^2), \qquad \mathcal{A}(O) = 1.$$

The formal series satisfies

$$G(w) = 1 - \frac{2}{5}w - \frac{2}{15}w^2 + O(w^3), \qquad g_j = \frac{4j-5}{4j+1}g_{j-1}.$$

The coefficient bound and full-tail proof in [4, Lemma 3.1] give $\mathcal{A} \equiv 1 \pmod{p^{4r}}$, $G_2/\mathcal{A} \equiv 1 \pmod{p^{12r}}$, and $1/\mathcal{A} \equiv 1 \pmod{p^{4r}}$. These analytic units equal one at O, so zero numerator points require no quotient $\log_E(O)/z(O)$.

The integral group identities

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i - n_i)]V_0$$

kill the torsion component of H and yield full logarithm ratios $r_i = u_i/h$ and $r_i' = (u_i - n_i)/(h-1)$. At the odd prime, $h, h-1$ are integral and at least one is a unit. Hence at least one full ratio is integral. Multiplication by the analytic units preserves the corresponding approximate valuation. The positive-order cross test transfers integrality to the other approximate ratio and thus to both full ratios.

Only now use $L_i = z(U_i)/z(H)$ integral. The H test gives $v(z(H)) \geq 3$, so every nonzero U_i has depth at least three. The primary errors have order at least twelve. The alternate errors have order at least four for $i = 1$ and twelve for $i > 1$. The integral full ratios convert these relative bounds into absolute congruences. Multiplication by $h, h-1$ and subtraction then give

$$L_1 \equiv n \pmod{p^3}, \qquad L_i \equiv n_i \pmod{p^{12}} \quad (i > 1).$$

This argument includes zero U_i or V_i, half-integral h, and inert primes. At the inert primes conjugation preserves the valuation.

5 Global soundness for sufficiently large parameters

The retained scalar circuit gives, for $c = |a - 1|$,

$$c \geq 80989^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \leq \mathcal{H} c^{19/24}. \tag{6}$$

These statements, including their integral group identities and exceptional-point exclusions, are the scalar conclusions of [2]; no comparison logarithm is used to establish them. Their circuit is retained, including the actual positive contact product r and the normalizer

$$r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \qquad R_s = \frac{A_N r}{1 + 80989^2 r}.$$

Let $\Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1$. The local comparison shows that each twelfth power of a comparison radical divides Δ. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (5) gives

$$B^2 \leq C 14^{19/48} B^{19/12} + M_2.$$

Since $19/12 < 2$, this bounds B by a fixed constant. Equation (6) gives $B \geq c - 1 \to \infty$ uniformly as $N \to \infty$, a contradiction. Thus $\Delta = 0$ eventually. The retained integral Lucas descent then gives

$$\log B > |a| - 3 \geq c - 4.$$

If $n - \beta \neq 0$, let J be its nonzero reduced numerator. The local input comparisons make both cubed radicals divide J, while $|J| \leq 2B \exp(h(\beta))$. For $B^2 \geq 2M_2$, (5) gives

$$\log B \leq \frac{12}{5}\log(2C) + \frac{19}{5}\log 2 + \frac{19}{5}h(\beta) \leq C' + \frac{19}{5}\mathcal{H} c^{19/24}.$$

The right side is $o(c)$, contradicting the preceding lower bound. Hence $n = \beta \in \mathbb{Z}$. If $X = u/v$ is reduced with $v > 0$, then $v^2 \mid 51$, and squarefreeness of 51 gives $v = 1$. All bounds depend only on fixed constants. Since L_N is increasing, a single N_0 works for every larger N. Its dependence on the aggregate additive constant is ineffective.

6 Simultaneous completeness

Fix an integer input and the coefficient parameter. Retain the scalar and Lucas construction of the source, and choose the integer labels, root points, two scales, and compact real and finite-place neighborhoods before selecting primes. The neighborhoods avoid every retained rational denominator; Lemma 3.1 ensures that the new denominator introduces none. Every local test then has a fixed lower valuation bound at each prime in the finite preparation set.

Choose a fixed positive integer D_0 clearing all contact coefficients $D_0^2\theta/a_*$ and sufficiently divisible at the prepared primes, including Σ. In the prime pattern write

$$\lambda = \frac{u}{D_0^2 w^2}.$$

The zero contact makes u one of the selected primes, outside that finite set. Thus $v_p(\lambda) \leq -2v_p(D_0)$ there. Make this upper bound low enough that $t_* = a_*\lambda$ has valuation below zero and all nonzero contacts. The exact valuation formulas are then

$$v_p(F_4(t_*)) = 20v_p(t_*), \quad v_p(e) = 20v_p(a_*), \quad v_p(q) = 18v_p(a_*) - 2v_p(\lambda).$$

The last expression becomes uniformly arbitrarily large. All new weighted test terms are therefore small simultaneously at the fixed prepared places. This is a fixed denominator dilation of the same finite prime pattern, allowed by [7].

Select the enlarged finite set of contact primes with the inherited opposite-order separation. The common-index congruences may retain primary depth twenty for H and the U_i; they are stronger than needed here. At the opposite places the points are formal. The new Taylor approximations give comparison precision twelve, and the constant approximation at V_1 gives precision four. Thus every nonconstant term of every factor has positive valuation at the chosen primary place, where $v_p(R) = 1$. There $W \equiv 1$. At the opposite place, the valuation-class argument gives either positive valuation for W^{-1} or residue one. Its trace is consequently a unit with residue one or two, including the inert case. Hence the pole input has valuation one at every selected contact prime.

The exact prime-pattern identity is

$$R = \frac{\prod_{\theta \in \Theta_4} (u - (\theta/a_*)D_0^2 w^2)}{(D_0^2 w^2)^{20} \delta^{22}}.$$

Its denominator is a square, so at every prime outside the selected prime numerators its valuation is even. Division by the squared trace preserves this parity. This argument includes primes dividing δ; no unit assertion about δ is made away from contacts. The chosen congruences give the local squareclasses at $2, 3, 17$, and the preparatory divisibility gives the required valuation bounds.

On the fixed compact real neighborhoods,

$$R \sim a_*^{-22}\lambda^{-2}, \qquad q \sim a_*^{18}\lambda^{-2}, \qquad W \longrightarrow 1, \qquad \mathrm{Tr}(W^{-1}) \longrightarrow 2.$$

All test powers in the factors are even, so the pole input tends to zero through positive values. It eventually lies in $(0, 17)$. The simultaneous selection supplies the required real size after all fixed data have been chosen. The pole-realization input supplies its witnesses, and the exact power identities fill the quadratic circuit. This proves completeness with no additional condition on N.

7 Exact quadratic specification

The weights are given by seven scalar quadratic rows

$$q_2 = q^2, \quad q_4 = q_2^2, \quad cq_4 = q, \quad q_8 = q_4^2, \quad q_5 = qq_4, \quad q_{10} = q_5^2, \quad cq_{23} = q_{10}^2.$$

At every contact $q \neq 0$, so these are the indicated powers, including $c = q^{-3}$. Their zero stratum is covered by (4).

Every degree from zero through twenty is a sum of two exponents in

$$\{0, 1, 2, 4, 5, 8, 9, 10\}.$$

Successive quadratic rows compute $t_*^2, t_*^4, t_*^5, t_*^8, t_*^9, t_*^{10}$. The coefficient table then expresses $F_4(t_*)$ as a quadratic polynomial in this basis. At the base scale $a_* = 1$, the same basis expresses δ^{20}, so no separate numerator power chain is needed. At the other scale, the existing values $x = a_*\delta, x^2, x^4$ are extended by $x^5 = xx^4$ and $x^{10} = (x^5)^2$, and the first equation of (4) uses $(x^{10})^2$. The existing trace-clearing coordinate now represents $e\delta^2$.

The high-point approximants G_2 are affine expressions in the retained f^2 and w^2 coordinates. They introduce no additional field value. At V_1 the numerator is simply $z(V_1)$. The shared new field powers are

$$C_1^4, \ A_3^4, \ A_4^4, \ A_4^8,$$

each with two rational coordinates. The fourth powers of B_0, B_2, B_3, B_4 already occur in the Taylor circuits. Each detector has five field factors and three private intermediate products, using sixteen coordinates; the final product uses the retained W coordinates. The count is

retained coordinates and contact/logarithm circuits	239
four shared field powers	8
five factors and intermediate products for two detectors	32
total coordinates	279
retained and contact/logarithm quadratic rows	209
shared power rows	8
factor and product rows	36
total quadratic rows	253

The producer clears denominators and primitive-normalizes each row before squaring. The independent checker imports no producer. It reconstructs the degree-twenty contact polynomial by forward differences of twenty-one exact evaluations, checks all twenty roots and the full quadratic basis, reconstructs every changed row and every affine Taylor substitution, verifies all retained rows, and independently expands the whole sum of squares by unordered pairs. It separately checks the δ^2 trace normalization, the global G_2 nonvanishing certificate, and 70912 supplementary valuation cases. The infinite arithmetic arguments are the proofs in the preceding sections.

The expanded quartic has 3693 terms before powers of A_N belonging to the same coordinate monomial are collected, and 3692 monomials after collection. Its unique coefficient involving A_N^2 is $A_N^2 + 1$, at the square of the actual scalar contact-product coordinate. Every other coefficient is a nonzero constant or a nonzero constant multiple of A_N. Consequently no support cancellation occurs for positive A_N. The largest constant coefficient C_0 has 1520 bits, the largest linear multiplier is 13118436242, and exact comparison gives

$$C_0 > 13118436242^2, \qquad 80989^{90} + 1 < C_0 < 80989^{94} + 1.$$

Thus $H(P_N) = \max(C_0, A_N^2 + 1)$ for every $N \geq 1$ and the asserted exact height holds for $N \geq 23$. This is a coefficient statement; it does not make the defining threshold effective.

References

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[3] Project repository, *Common local weights and quartics in 297 and 342 variables*, `shared_local_weights_20261109_v12.tex`.

[4] Project repository, *An explicit quartic in 322 variables*, `quartic322_20261109_v13.tex`.

[5] Project repository, *Height Bounds from Disjoint Sets of Rational Contacts*, `contact_height_amplification_20261109_v13.tex`.

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